Luwei Yang

Physical Oceanographer, Postdoctoral Research Fellow — ANU profile; Scopus;
Research School of Earth Sciences, Australian National University — luwei.yang@anu.edu.au

Education

2015–2019 Ph.D. in Quantitative Marine Science, University of Tasmania, Hobart, Australia.

2011–2015 B.Sc. in Marine Science, Ocean University of China, Qingdao, China.

Professional Experience

2023–present Postdoctoral Research Fellow, Research School of Earth Sciences, Australian National University, Canberra, ACT, Australia.

2020–2023 Postdoctoral researcher, Department of Atmospheric & Oceanic Sciences, UCLA, Los Angeles, California, United States.

Publications

[8] **Yang, L.**, Shakespeare, C.J., B.K. Arbic: Propagating internal-tide-induced wave stresses resolve discrepancies in ocean tide energetics, submitted.

[7] **Yang, L.**, C.J. Shakespeare, A.K. Morrison, A.M. Hogg, A.H. Gibson, and B.K. Arbic, 2025: Improving M_2 barotropic tide solutions: sensitivity to model resolution and nonlinear wave dynamics. *Journal of Advances in Modeling Earth Systems*, under review (preprint available in ESS Open Archive, doi: 10.22541/essoar.176005616.66140750/v1).

[6] Barnes, A.J., N.C. Constantinou, A.H. Gibson, A.E. Kiss, C. Chapman, J. Reilly, D. Bhagtani, **L. Yang**, 2024: regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6. *Journal of Open Source Software*, **9(100)**, doi: 10.21105/joss.06857

[5] **Yang, L.**, R. Barkan, K. Srinivasan, J.C. McWilliams, C.J. Shakespeare, and A.H. Gibson, 2023: Oceanic eddies induce a rapid formation of an internal wave continuum. *Communications Earth & Environment*, **4(1)**, doi: 10.1038/s43247-023-01137-1

[4] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2023: Lee waves break eddy saturation of the Antarctic Circumpolar Current. *Geophysical Research Letters*, **50(11)**, doi: 10.1029/2023GL103866.

[3] Barkan R., K. Srinivasan, **L. Yang**, J.C. McWilliams, J. Gula, and C. Vic, 2021: Oceanic mesoscale eddy depletion catalyzed by internal waves. *Geophysical Research Letters*, **48(18)**, doi: 10.1029/2021GL094376.

[2] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2021: The impact of lee waves on the Southern Ocean circulation. *Journal of Physical Oceanography*, **51(9)**, 2933-2950, doi: 10.1175/JPO-D-20-0263.1.

[1] **Yang, L.**, M. Nikurashin, A.M. Hogg, and B.M. Sloyan, 2018: Energy Loss from Transient Eddies due to Lee Wave Generation in the Southern Ocean. *Journal of Physical Oceanogrphy*, **48(12)**, 2867–2885, doi: 10.1175/JPO-D-18-0077.1.

Presentations

Invited talks

[4] **Yang, L.** Eddy-wave interactions in the ocean. *Fudan-Guanghua International Forum for Young Scholars, Subforum of Symposium on Frontiers in Atmospheric and Oceanic Sciences, Fudan University*, Shanghai, China, 18 February 2023.

[3] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress. *OCES Departmental Seminar, The Hong Kong University of Science and Technology*, Zoom, 30 September 2022.

[2] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress. *Physical Oceanography Dissertations Symposium (PODS) XI*, Lihue, Kaua'i, Hawaii, United States, 17-21 October 2021.

[1] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the Southern Ocean circulation and its response to changes in wind stress. *Polar Oceans seminar, British Antarctic Survey*, Zoom, 4 November 2020.

Selected Oral Presentations

[15] **Yang, L.**, C. J. Shakespeare, A. K. Morrison, A. McC. Hogg, A. H. Gibson,and Brian Arbic. Implementing and evaluating a novel tidal wave drag parameterization in MOM6. *Ocean Modelling and Observations Workshop*, Canberra, ACT, Australia, 2-4 July 2024.

[14] **Yang, L.**, C. J. Shakespeare, A. K. Morrison, A. McC. Hogg, A. H. Gibson,and Brian Arbic. Connecting ocean tides to the large-scale ocean circulation: Tidal parameterization development in MOM6. *Bluelink Annual Science Meeting*, Melbourne, Victoria, Australia, 19-20 March 2024.

[13] **Yang, L.**, R. Barkan, K. Srinivasan, J. C. McWilliams, C. J. Shakespeare, A. H. Gibson. Oceanic eddies induce a rapid formation of an internal wave continuum. CASPO seminar, Scripps Institution of Oceanography, La Jolla, California, United States, 28 February 2024.

[12] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. Lee waves break eddy saturation of the Antarctic Circumpolar Current. *Ocean Sciences Meeting 2024*, HE41B-07, New Orleans, Louisiana, United States, 18-23 February 2024.

[11] **Yang, L.**, R. Barkan, K. Srinivasan, J. C. McWilliams, C. J. Shakespeare, A. H. Gibson. Oceanic eddies induce a rapid formation of an internal wave continuum. *Ocean Sciences Meeting 2024*, PS21A-07, New Orleans, Louisiana, United States, 18-23 February 2024.

[10] **Yang, L.**, R. Barkan, K. Srinivasan, J. C. McWilliams, C. J. Shakespeare, A. H. Gibson. Regulation of internal wave energy and its dissipation by eddies. *MEL Seminar, State Key Laboratory of Marine Environmental Science (Xiamen University)*, 22 February 2023.

[9] **Yang, L.**, R. Barkan, K. Srinivasan, and J. C. McWilliams. On the propagation of wind-generated near-inertial waves: the importance of submesoscale currents. *Ocean Sciences Meeting 2022, PS11, The dynamics of interacting internal waves and (sub)mesoscale flows*, Virtual, 28 February - 4 March 2022.

[8] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The impact of lee waves on the response of the Southern Ocean circulation to changes in wind stress. *Second California Geophysical Fluid Dynamics (CalGFD) Meeting*, Zoom, 20-21 August 2020.

[7] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. Impacts of Lee Waves on the Southern Ocean Circulation and its Sensitivity to Wind Stress. *22nd Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD)*, Idealized Model Approaches to the Atmosphere and Ocean Circulation 7.2, Portland, Maine, United States, 24-28 June 2019.

[6] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Impacts of Bottom Drag on the Sensitivity of the Southern Ocean Circulation to the Changing Wind. *2018 Consortium for Ocean Sea Ice Modelling in Australia (COSIMA) workshop*, Canberra, Australia, 7-8 May 2018.

[5] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *2018 Ocean Sciences Meeting*, PO12A-04, Portland, Oregon, United States, 11-16 February 2018.

[4] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Impacts of Bottom Drag on the Sensitivity of the Southern Ocean Circulation to the Changing Wind. *Joint AMOS National Conference and the International Conference on Southern Hemisphere Meteorology and Oceanography (AMOS-ICSHMO 2018)*, Sydney, Australia, 5-9 February 2018.

[3] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *Australian Research Council's Centre of Excellence for Climate System Science (ARCCSS) 2017 Workshop*, Canberra, Australia, 30 October - 2 November 2017.

[2] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *21st Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD) and the 19th Conference on Middle Atmosphere*, Mesoscale Ocean Dynamics 13.2, Portland, Oregon, United States, 26-30 June 2017.

[1] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. The Role of Lee Waves for Dissipation of Transient Eddies in the Southern Ocean. *AMOS/MSNZ Conference & ANZ Climate Forum 2017*, Canberra, Australia, 7-10 February 2017.

Selected Poster Presentations

[3] **Yang, L.**, C. J. Shakespeare, A. K. Morrison, A. McC. Hogg, A. H. Gibson, and Brian Arbic. Implementing and evaluating a novel tidal wave drag parameterization in MOM6. *Gordon Research Conference on Ocean Mixing*, South Hadley, Massachusetts, United States, 9-13 June 2024.

[2] **Yang, L.**, R. Barkan, K. Srinivasan, and J. C. McWilliams. The energetics of wind-generated near-inertial waves. *23rd Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD)*, Breckenridge, Colorado, United States, 13-17 June 2022.

[1] **Yang, L.**, M. Nikurashin, A. M. Hogg, and B. M. Sloyan. Impacts of Lee Waves on the Southern Ocean Circulation and its Sensitivity to Wind Stress. *Ocean Sciences Meeting 2020*, PS14A-2817, San Diego, California, United States, 16-21 February 2020.

Selected Honours and Awards

- Best Early Career Researcher (ECR) presentation, Ocean Modelling and Observations Workshop, Canberra, ACT, Australia, 2024.
- Travel grant, Physical Oceanography Dissertations Symposium (PODS) XI, Lihue, Kaua'i, Hawaii, United States, 2021.
- Bursary (€1000), 2017 Summer School on the Fluid Dynamics of Sustainability and the Environment (FDSE), Ecole Polytechnique, Palaiseau, France, 2017.
- Student Oral Presentation Honourable Mention, AMOS/MSNZ Conference & ANZ Climate Forum, Canberra, Australia, 2017.
- Excellent Graduates in the Class of 2015, Ocean University of China, Qingdao, China, 2015.
- National Scholarship, Ministry of Education of China, 2014.
- The Oustanding Student Award, Ocean University of China, Qingdao, China, 2012, 2014.
- The First Class Scholarship Award, Ocean University of China, Qingdao, China, 2014.

Teaching Experience

- Lecturing: Fundamentals of Climate Science (EMSC2021), Australian National University. Semester 2, 2024. Convenor: Dr. Callum Shakespeare.

- Marking: Dynamical Oceanography (KSM310), University of Tasmania. 2017-2019. Coordinators: Dr. Maxim Nikurashin, Prof. Neil Holbrook, Prof. Nathan Bindoff.
- Marking: Advanced Oceanography (KSA306), University of Tasmania. 2016. Coordinator: Dr. Maxim Nikurashin
- Demonstrating and marking: Our Changing Climate (KGA320), University of Tasmania. 2016. Coordinator: Prof. Neil Holbrook

Field Experience

- *CTD watch stander*, leg 2 of the GO-SHIP P15S Hydrographic Section, Wellington (NZ) - Lautoka (Fiji), South Pacific, Monitoring Ocean Change and Variability along 170°W from the ice edge to the equator, Chief Scientist: Susan Wijffels (CSIRO Oceans and Atmosphere, WHOI), R/V Investigator, May-June 2016 (35 days).

Supervision Experience

- Co-supervision: Anupiya Ellepola, Ph.D., Australian National University (2024-).
- Supervision: Audrey Smart, Ph.B., Australian National University (2024).

Professional Services

- *Collaborator*, ACCESS-OM3 development, 2025 - present.
- *Reviewer*, Journal of Physical Oceanography, Journal of Geophysical Research Oceans, Journal of Fluid Mechanics, Journal of Advances in Modeling Earth Systems, Geophysical Research Letters.
- *Group meeting coordinator*, Climate and Fluid Physics, Research School of Earth Sciences, College of Science, ANU, Januray 2024 - present.
- *Discussion leader*, Gordon Research Seminar on Ocean Mixing, South Hadley, Massachusetts, United States, June 2024.
- *Postdoctoral Fellow Selection Committee Member*, Australian National University, March 2024.
- *UTAS PhD student representative* and *Early Career Researcher (ECR) Committee member*, ARC Centre of Excellence for Climate Extremes, 2018.
- *Session Chair*, ARC Centre of Excellence for Climate Extremes Annual Workshop, November 2018.
- *Volunteer*, 22nd Conference on Atmospheric and Oceanic Fluid Dynamics (AOFD), June 2018.

Outreach

- *Volunteer*, ANU open day, 29 March 2025
  - Demonstrated and engaged in a lab experiment with prospective students and their parents on 'Icebergs in the Southern Ocean' using a rotating tank (Weather in a Tank Apparatus) and a thermal camera.
- *Volunteer*, IMAS Work Experience program, June-July 2018.
  - Designed and delivered an 1-hour lecture on 'How do we study oceans? - Research in Physical Oceanography' for two groups of high school students across Australia;
  - Demonstrated and engaged in a lab experiment with high school students on 'Ocean gyres (Western Intensification phenomenon)' using a rotating tank (Weather in a Tank Apparatus).
- *Volunteer*, Marine Debris Clean-ups at Charlotte Cove, Tasmania, Australia, 19 May 2018.
  - Clean-up and categorization of marine plastics from local beaches;
  - D'Entrecasteaux and Huon Collaboration project;
  - Organizer: NRM South, South Hobart, Tasmania, Australia.

- *Volunteer*, Young Antarctic Scientists program, May and December 2018.
  - Technical session, 13 December. Engaged in an outdoor practical session, in which year 4 students at Clarendon Vale Primary School observed and learned how to fly a drone.
  - Biogeochemical oceanography session, 13 December. Coordinated a group activity in which year 4 students at Clarendon Vale Primary School used baking soda and vinegar to blow the balloons with CO_2.
  - Physical Oceanography session, 16 May. Engaged in a lecture and an experiment, in which oil, water, honey, vinegar are slowly added into and settled in a tube, aiming to deliver the idea that 'ocean has layers', with year 4 students at Clarendon Vale Primary School.

Last updated: December 29, 2025